Mohamed Elsawah

Software Engineer | Full-Stack Developer | AI Automation Engineer
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PROFESSIONAL SUMMARY

Software engineer and founder building production AI products and enterprise systems end to end. Owns product, architecture, backend, frontend and delivery on multi-tenant ERP platforms, omnichannel messaging infrastructure, sales operations systems and AI-assisted clinical software. Previously built systems handling 700,000+ annual transactions and 20+ automation workflows. Core stack: Next.js, React, .NET, PostgreSQL, React Native, n8n and OpenAI.

CORE COMPETENCIES & TECHNICAL SKILLS

Languages	C/C++, C# (.NET), Python, JavaScript / TypeScript
Web & Full-Stack	Next.js, React, React Native / Expo, ASP.NET Core MVC/API, Tailwind CSS, Redux Toolkit, RTL / bilingual (Arabic-English) UI
AI & Automation	OpenAI, RAG Knowledge Bases, AI assistants with tool use, ElevenLabs, n8n, Python scripting, Webhooks, Playwright
Integrations	Meta WhatsApp Cloud API, Messenger & Instagram Messaging, Chatwoot API, Meta Graph API / OAuth, Firebase, SuiteCRM
Databases & Infra	PostgreSQL, Prisma, EF Core, SQL Server, Supabase, Docker, AWS / VPS, Vercel, Git
Architecture	Clean Architecture, multi-tenant systems, RESTful APIs, JWT / RBAC, SOLID Principles

PROFESSIONAL EXPERIENCE

Founder | KemexTech LLC | Remote — Jun 2026 – Present

- Founded a Wyoming-registered software and digital-services company, currently at an early stage and targeting small businesses in the US.
- Designed, built and deployed the company website and portfolio (Next.js on Vercel).

Co-Founder & Technical Lead | Jisrmed AI | Sheikh Zayed, Giza (Hybrid) — May 2026 – Present

- Own the entire technical side of JISR AI, a clinical workspace for outpatient clinics currently at MVP stage: data model, interview engine, AI layer, doctor report, patient experience and clinic dashboard.
- Built the pre-visit interview where patients complete their history from their phone, so the doctor starts the visit with a structured file instead of paper intake.

Software & AI Automation Engineer | PURA Egypt | Cairo (Hybrid, Part-Time Contract) — Mar 2026 – Present

- Technical execution lead on ERP, CRM and automation products for Egyptian companies, including a construction ERP, a sales CRM and an omnichannel inbox (see Key Projects).
- Deliver end to end: product scoping, architecture, backend, bilingual Arabic/English frontend, deployment and handover.

AI & Automation Engineer | Dokkan Agency | Tanta — Oct 2025 – Apr 2026

- Engineered a centralized decision-making dashboard integrating data from Meta, TikTok, Snapchat and Google Ads.
- Developed 20+ automated workflows with n8n and Python to calculate real-time ROAS and ROI.
- Built scraping bots for Facebook Ads Library and Google Maps to extract and qualify B2B leads; deployed a Discord Sales Reporting Bot and a Product Testing Bot that cut manual QA time.

Lead Automation & Generation Specialist | Aptelocome Agency (Orange Business Partner) | Remote — Jul 2025 – Sep 2025

- Used automated data collection and workflows to streamline B2B lead generation and qualify leads at scale, replacing manual research.

Freelance Full-Stack .NET Engineer | Self-Employed | Remote — Dec 2024 – May 2025

- Designed a .NET Core MVC backend handling 700,000+ annual transactions with optimized SQL queries.
- Built a custom QR generation system integrated with warehouse scanners for real-time tracking and inventory management.

KEY PROJECTS & ENGINEERING DELIVERABLES

Construction ERP — Operating System for Finishing Contractors (PURA) — *Full ownership: product to delivery*

- Multi-project platform linking tender and BOQ, contract, change orders, client certificates, subcontractors, cost and cash in one source of truth, with approval workflows before execution.

- Covers site diaries, procurement and inventory (RFQ comparison, PO, stock movements), project accounting (chart of accounts, journals, bank reconciliation) and site-based HR (geofenced attendance, leave, payroll).
- Built an in-product AI assistant that reads live data through role-restricted tools, requires confirmation before any write and logs every action to an audit trail.

  Stack: Next.js 16, React 19, TypeScript, .NET 10, ASP.NET Core, Clean Architecture, PostgreSQL, EF Core, Docker on AWS

JISR AI — Clinical Workspace for Outpatient Clinics — *Co-Founder / Technical Lead (MVP stage)*

- Own the full technical side: data model, interview engine, AI layer, doctor report, patient experience and clinic dashboard.
- Pre-visit interview (link + QR, Arabic/English, voice-enabled questions) whose path adapts to the patient's answers and visit type, producing a structured doctor-facing report with flagged findings and suggested tests. The AI organizes and presents; it does not diagnose or prescribe.
- Returning-patient flow loads only what exists in the record and shows the source of every prior value. Specialty logic is a configuration file; OB/GYN is the first protocol.

  Stack: Next.js 16, React 19, TypeScript, PostgreSQL, Prisma, OpenAI, ElevenLabs

Omnichannel Inbox with AI Agent (PURA / KemexTech) — *Integration, Deployment & Operations*

- Integrated WhatsApp, Messenger, Instagram and live chat into a single self-hosted inbox (Chatwoot), so all customer messages land in one place and replies return on the original channel.
- Handled the Meta side end to end: business verification, WhatsApp Business API, Messenger and Instagram integrations, and Technical Provider onboarding.
- Deployed the platform on our own servers and manage and monitor it in production. Added an AI agent on top for automated first-line replies, with automation rules for greetings, labels and routing, plus a React Native agent app.

  Stack: Chatwoot, Meta Cloud API, Messenger & Instagram Messaging, Docker / VPS, React Native, Expo, Firebase

Sales CRM — Lead Intake to Close (PURA) — *Full ownership*

- Lead capture via Meta Lead Ads webhooks and a generic webhook for TikTok, Google, LinkedIn, Zapier/Make; duplicate prevention by phone or external ID.
- Round-robin assignment to the least-loaded rep, follow-up stages (Interested 7/15/30, Callback, Won/Lost) with a delay list, bulk Excel import up to 1,000 rows, masked phone numbers and role-based access with audit logging.

  Stack: Next.js 16, React 19, TypeScript, Supabase Postgres, NextAuth, Meta Graph API, Vercel

Textile Factory Management System — *Full-Stack AI Engineer*

- Multi-tenant ERP for textile manufacturing covering production planning, resource scheduling, timeline tracking and cost estimation across multiple factory tenants.

Autonomous AI Sales Agent — *AI Automation Engineer*

- n8n + OpenAI agent qualifying B2B leads 24/7 on WhatsApp: RAG-based FAQs, purchase-intent detection and lead routing into SuiteCRM. Demo

Loyalty & Rewards Module (ERP) — *Full-Stack Engineer*

- Loyalty module inside a larger ERP: dynamic points tracking, tier management and automated customer engagement workflows. Demo

EDUCATION & TRAINING

B.Sc. in Computer Engineering | Tanta University, Egypt — Expected 2027
Professional Diploma, .NET Web Development | Digital Egypt Pioneers Initiative (DEPI) — May 2024 – Oct 2024
Entrepreneurship & Startup Fundamentals | Kenko Ventures Pre-Incubation Program — May 2025 – Oct 2025

ACTIVITIES & ACHIEVEMENTS

- **ECPC Participant:** Egyptian Collegiate Programming Contest. **Competitive Programming:** active on Codeforces and LeetCode.
- **Volunteering:** Team Leader at Life Makers Association.